

Human Proteostasis Network

GO Mapping Review

Manual PN-to-GO propagation with explicit curation status

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Source And Scope

Current completed pass:

- Human Proteostasis Network Annotation 4.3.11
- Release date: 2026-04-17
- Local schema and mappings are keyed to this release

Why Manual Mapping Is Required

The PN workbook is a role taxonomy, not a GO annotation table.

PN rows can mean:

- direct molecular activity
- cellular component membership
- pathway-stage context
- family or domain metadata
- broad proteostasis systems membership

The mapping cycle separates PN membership from GO assertion.

Coverage Completion

Every 2026 PN source code is now accounted for.

Total nodes: 2029. Leaf nodes: 1348.

Level	Total	Pending	Mapped	Context	No map	Deferred	Missing
Branch	9	0	0	1	8	0	0
Class	42	0	9	16	17	0	0
Group	297	0	133	31	133	0	0
Type	800	0	233	26	541	0	0
Subtype	881	0	105	16	760	0	0

Inventory: 2029 subject curation records, one per PN node.

Curation Status Policy

Every PN node has a curator-facing status.

- pending review: temporary state during a new PN release mapping pass
- mapped: reviewed and mapped to GO
- context only: GO relationship recorded, but unsafe to propagate
- no mapping: reviewed, no GO mapping should be made
- deferred: reviewed, but blocked by evidence, taxonomy ambiguity, or a term gap

Current curation inventory:

- 480 mapped
- 90 context only
- 1459 no mapping
- 0 pending review

Projection Output

Projection through current propagating mappings produced 3565 unique gene-GO pairs.

GOA source: `~/repos/go-db/db/goa_human.ddb`

Status	Count
already in GOA exactly	1928
entailed by GOA closure	512
more specific than existing GOA	305
supported by GOA regulation	35
new to GOA	753
no local GOA	32

Only the 1093 candidate additions enter manual rereview queues.

Extra Scrutiny Audit

The current mapping set is intentionally conservative, but 430 GO-bearing records are flagged as requiring gene-level review before changing AIGR YAML.

Top flag types:

- 313 contextual or regulatory source labels
- 262 domain or family metadata labels
- 190 branch/class-level mappings
- 131 broad or context-losing GO targets
- 89 `too_broad_to_propagate` mappings excluded from projections

Report: `reports/pn_mapping_audit/unusual_propagations.tsv`

Propagation Lessons

Projection labels are queue labels, not curation decisions.

Examples from rereview:

- **BCAP31** : accepted ERAD pathway projection
- **EDF1** : accepted RQC/PQC term, rejected broad translation terms
- **TOMM20** : rejected generic protein-import projection as broader than existing mitochondrial import annotation
- **HSPA8** : rejected aggrephagy projection in favor of better-supported CMA
- **RAB7A** : rejected fusion term where evidence better supported post-fusion maturation

Risk Areas

Use extra scrutiny for:

- broad Translation branch/class propagation
- mitochondrial protein import buckets
- ALP docking/fusion and maturation labels
- CMA versus aggrephagy boundaries
- UPS ubiquitin/UBL-binding context buckets
- regulator/modulator labels that point to pathway process terms

These should drive expert review, not automatic gene-review edits.

PN-GO Bridge Product

The deliverable should be a GO-ready companion layer for each PN release.

Each PN row gets:

- GO bridge status: direct annotation, review first, ontology gap, PN context only
- evidence basis: primary literature, review, domain/family, orthology, database seed
- directness: direct gene evidence, family inference, module context, regulator, hypothesis
- candidate GO term, if one exists
- gene exceptions that must not inherit the row-level mapping
- ontology-gap status: covered, NTR candidate, NTR justified, design pattern, GO-CAM/extension

Outputs: GO candidate table, exception table, ontology-gap list, PN feedback table.

Current Artifacts

Primary files:

- `projects/PROTEOSTASIS/mappings/*.yaml`
- `projects/PROTEOSTASIS/reports/pn_mapping_coverage/`
- `projects/PROTEOSTASIS/reports/pn_projection/`
- `projects/PROTEOSTASIS/reports/pn_mapping_audit/`
- `projects/PROTEOSTASIS/reports/pn_mappings/pn_mappings.xlsx`
- `projects/PROTEOSTASIS/reports/pn_taxonomy_tree/`

HTML page: `pages/projects/PROTEOSTASIS.html`

Browser: `pages/projects/PROTEOSTASIS-tree.html`

Next Work

1. Work the 1093 candidate additions as manual AI GR rereview tasks.
2. Use unusual-propagation audit as a guardrail.
3. Materialize the PN-GO bridge fields for every mapped/context PN row.
4. Promote only evidence-backed gene-level decisions to review YAML.